

Class Test -II (JAN. - 2023)
2AMRC1- Applied Mathematics -II
(B.E. Iyr ETC A, B)

Time: 70 min.

Maximum Marks: 20

Note: Attempt all four questions. Questions must be solved at one place. Each step should be clear and well defined.

Q.1 Solve the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + 2y = \sinh x + \sin\sqrt{2}x$. 05

Q.2 Solve the differential equation $x^2 \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} + 5y = x \log x$. 05

Q.3 Form the Partial differential equations 05

(i) $x^2 + y^2 + (z - a)^2 = b^2$ (ii) $f(x + y + z, x^2 + y^2 + z^2) = 0$.

Q.4 Find the general solution of $x(z^2 - y^2) \frac{\partial z}{\partial x} + y(x^2 - z^2) \frac{\partial z}{\partial y} = z(y^2 - x^2)$. 05